CNS Blocking Error detection

Consider the value of $k=2$ & $n=3$

data word - 2 bit Code word - 3 bit

dw	cw
00	000
01	011
10	101
11	110

XOR gate

000 ⊕ 011 → 011 → 2 1's

000 ⊕ 101 → 101 → 2 1's

000 ⊕ 110 → 110 → 2

011 ⊕ 101 → 110 → 2

011 ⊕ 110 → 101 → 2

101 ⊕ 110 → 011 → 2

dmin = 2

Consider $k=3$ $n=4$

000	0000
001	0011
010	0101
011	0110
100	1001
101	1010
110	1100
111	1111

Research Methodology

Literature Review and Technical Reading

1.1 → 1. Literature Review
2. Technical Reading

1.2 → 1. Existing Knowledge
2. New Knowledge

1.3 → 1. Analysis and Synthesis of prior art
2. Bibliography database – IEEE plore, science direct, scopes springer, Elsevier
3. web of science 4. Google & Google scholor
5. Effective Search

The way forward

1. Introduction to Technical reading
2. Conceptualizing Research
3. Critical and Creative Reading
4. Taking notes while Reading
5. Reading Mathematics and Algorithms
6. Reading a Datasheet



CNS

Q1. Find the CRC for the data 1001 with using the generator 1101

$C = (7,4)$

$K = 4$

Sol⁴ data ← 1001

Generator (divisor) ← 1101

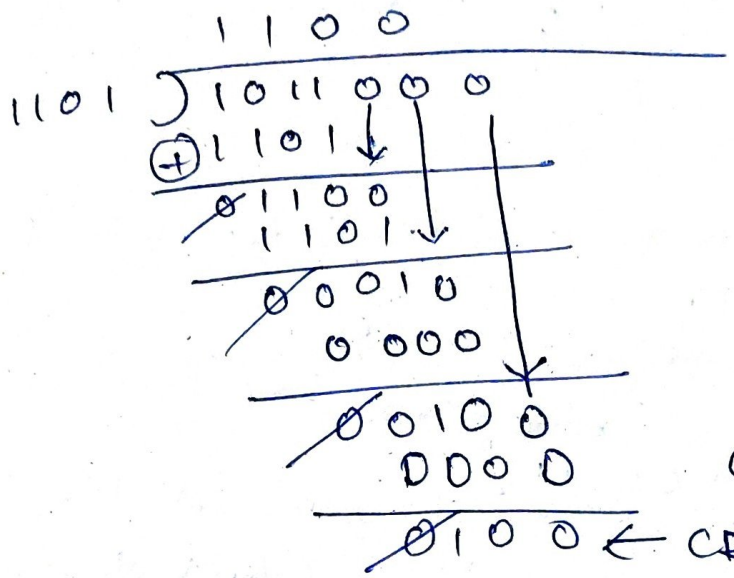
dataword ← 1001000 ⊕ → XOR

```
        1 1 1 1
      ---------
1101 ) 1 0 0 1 0 0 0
     ⊕ 1 1 0 1
     ---------
       0 1 0 0 0
       ⊕ 1 1 0 1
       ---------
         0 1 0 1 0
         ⊕ 1 1 0 1
         ---------
           0 1 1 1 0
           ⊕ 1 1 0 1
           ---------
             0 0 1 1
```

Codeword ⇒ 1001011

Q2 Given dataword 1011 Generator & 1101 find the CRC & codeword.

Ans data ← 1011000



Codeword → 1011100

Codeword = data + CRC

Q3 consider the data pattern 1001 Create word polynomial Generator 1101 ⟶ $p(x) = x^3+x^2+1$

→ 1001

$p(x) = 1.x^3 + 1.x^0 = x^3+1$ degree 3

Q4 for a dataword $x^5 + x^4 + x^2 + 1 = M(x)$

Ans $G(x) = x^3 + x + 1$ Generator Convert message & generator into bit pattern

Ans 110101 ← Message

1011 ← Generator

Leftshift $x << y$ $x * 2^y$

$x >> y$ $x / 2^y$

Q4 Consider data word 10011

Ans $x^4 + x + 1$

CRC division Polynomial

x^3+x+1) x^6 + x^3 (x^3+x

$x^6 + x^4 + x^3$

$x^4 + x^2 + x$

$x^2 + x$ → Reminder (CRC)

dataword - x^6+x^3

multiply by highest degree of Generator $x^3(x^3+1) = x^6+x^3$ ↓ dataword

Codeword = dataword + CRC

$= x^6+x^3+x^2+x$

1001110

In the binary add the no of zeros at the right part by considering the highest degree of the ~~divisor~~ Generator polynomial

a) data - 1101 generator = x^3+1 → 101

a) Consider the dataword - 1001 Generator ⇒ x^3+x+1

$(x^3+1) \times x^3$

x^6+x^3

x^3+x+1) x^6 + x^3

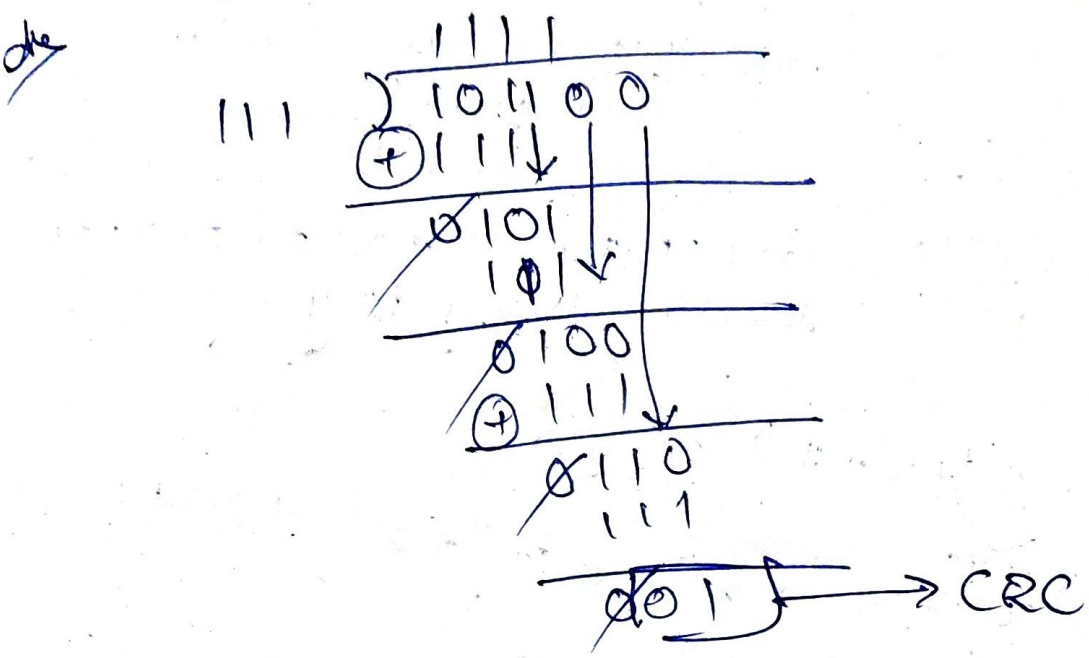
do this using binary because modulo 2 method.

↓

x^2+x

codeword =

Q) Find the CRC & transmitted codeword for dataword - 1011 Generator $G(x) = x^2+x+1$ (111) using modulo 2 method

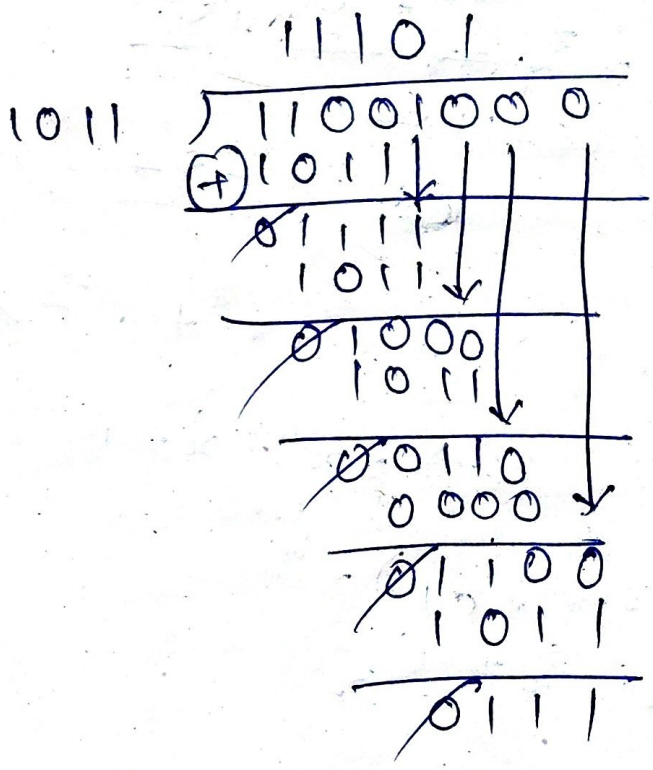


1011 01 = Codeword

data

Q) dataword - 11001 $G(x) = x^3+x+1$ (1011) CRC = ?

modulo 2



Codeword = 11001111

Continue assume that the recived codeword is 11001110 check wether the codeword is corrupted or not at the receiver

— Yes error as 1 bit is changed because Syndrome is 1

Q) data - 10101 Generator x^3+x^2+1 - 1101
Find CRC ξ Codeword using poly.
$_{4\,3\,2\,1\,0}$

→

$$(x^4+x^2+1)\,x^3 \rightarrow$$

$$x^3+x^2+1 \;\overline{)}\; x^7+x^5+x^3 \;(\; x^4+x^3+x+1$$

$$x^7+x^6+x^4$$

$$x^5+x^6+x^4+x^3$$

$$x^5+x^6+\quad+x^3$$

$x^4 \rightarrow$ CRC

$x^4+x^3+x \rightarrow$ CRC

x^3+x^2+1

x^3+x

Codeword → x^7 ... 1011100

$x^2+x+1 \rightarrow$ CRC

Codeword $- x^7+x^5+x^3+x^2+x+1$

Q) dataword 11001 $\quad G(x) = x^4+x+1$ poly
$_{5\,4\,3\,2\,1\,0}$

Ans $\quad (x^5+x^4+x^3+1)\,x^4$

$$x^9+x^8+x^7+x^4$$

x^4+x+1) $x^9+x^8+x^7+x^4$ ($x^5+x^4+x^3+x^2+1$

$x^9+x^6+x^5$

$x^8+x^7+x^6+x^5+x^4$

$x^8+\quad\quad+x^5+x^4$

x^7+x^6

$x^7+x^4+x^3$

$x^6+x^4+x^3$

$x^6+\quad+x^3+x^2$

x^4+x^2

$x^4+\quad x+1$

x^2+x+1

CRC ←

Codeword - $x^9+x^8+x^7+x^4+x^2+x+1$

determine codeword corrupt or not

x^4+x+1) $x^9+x^8+x^7+x^4+x^2+x+1$ ($x^5+x^4+x^3+x^2+$

$x^9+\quad+x^6+x^5$

$x^8+x^7+x^6+x^5+x^4+x^2+x+1$

$x^8\quad\quad+x^5+x+1$

$x^7+x^6+x^2+x+1$

$x^7+\quad+x^4+x^3$

$x^6+x^4+x^3+x^2+x+1$

$x^6+\quad+x^3+x^2$

x^4+x+1

x^4+x+1

000 → Syndrome 0

not corrupted

$M(x) = x^5 + x^4 + x^2 + 1$

$G(x) = x^3 + x + 1$

$$x^3+x+1 \;\overline{)\; x^8 + x^7 + x^6 + x^5 \;}\; (x^5 + x^4 + x^3 + x^2 + 1)$$

$x^8 + \quad + x^6 + x^5$

$x^7 + x^6 + x^3$

$x^7 + x^5 + x^4$

$x^6 + x^5 + x^4 + x^3$

$x^6 + \quad + x^4 + x^3$

x^5

$x^5 + x^3 + x^2$

$x^3 + x^2$

$x^3 + x + 1$

$x^2 + x + 1$

Codeword - $x^8 + x^7 + x^5 + x^3 + x^2 + x + 1$

1 1 0 1 0 1 1 1

RM

1.7 Impact of Title and keywords on citation, Knowledge flow through Citation.

1.8 Acknowledgements and attributions → Books, Dissertation, dedication.

(Q) (7 12 13 0 8) ← message (4 bit number) Calculate checksum at Senderside & new checksum at Racver side Calculate fore corruption

→

2	7
1	2
1	3
	8
4	0

→

32	16	8	4	2	1
1	0	1	0	0	0

↓ add

0 0 0 0 1

1000
10
1010 → 1's complement → 0101 → 5

Sender (7, 12, 13, 0, 8, 5)

Reciver 7 + 12 + 13 + 0 + 8 + 5 → 45

32 16 8 4 2 1
1 0 1 1 0 1
→ + 1 0
1 1 1 1

↓ 1's complement

0 0 0 0 → 0

(7, 12, 13, 0, 8, 0)

Calculate sum → 0
so not corrupted

Q. Manually simulate fletcher algo to calculate the checksum $(2B)_{16}$ $(3F)_{16}$ $(6A)_{16}$ & $(AF)_{16}$ also show the result in a weighted checksum. Also calculate first R then L

Initial value L=00 R=00

item 1 (2B) L = L + R = 0 + 2B = 2B R = R + D1 = 0 + 2B = 2B

item 2 (3F) L = 2B + 3F = 95 R = 2B + 3F = 6A

item 3 (6A) L = 95 + 6A = D4 R = 6A + 6A = D4

item 4 (AF) L = D4 + AF = 183 → 69 R = D4 + AF = 83

∴ checksum = L × 256 + R

= EC * 256 + 83

= EC * 100 + 83 = EC83

256 | 95 | 16 | 0.6

Adler Checksum modulo – 65,521

R – 1 at the beginning

following no $(FB)_{16}(FF)(EF)(AA)_{16}$

divided → $(FBFF)_{16}$ $(EFAA)_{16}$ ← divided

These are the data

$L = (L+R) \bmod 65,521$ $R = (R+D_i) \bmod 65,521$

$Checksum = L \times 65536 + R$

Initial value	$L = 00$	$R = 1$
	$L = L + R$	$R = R + D_i$
item1 FB	$L = 0 + FC$	$R = 1 + FB = 00FC$
	$= FC$	
item2 FF	$L = FC + 01FB$	$R = 00FC + FF$
	$= 02F7$	$= 01FB$
item3 EF	$L = 02F7 + 02EA$	$R = 01FB + EF$
	$= 05E1$	$= 02EA$
item4 AA	$L = 05E1 + 0394$	$R = 02EA + AA$
	$= 0975$	$= 0394$

$$Checksum = 0975 * \overbrace{65536}^{10,000} + 0394$$

$$= 975 0394$$

Station on a wireless ALOHA network are a max of 600 km apart

if signal speed 3×10^8 m/s T_P T_B

$\frac{600 \times 10^3}{3 \times 10^8} = 2 \times 10^{-3}$ s = 2 ms

ans $T_P = (600 \times 10^3)/(3 \times 10^8) = 2$ ms

$k = 2$ the range of R is $\{0, 1, 2, 3\}$

$T_B = R * T_P$

$R = 2^K - 1 = 2^2 - 1 = 3 \rightarrow 0\ 1\ 2\ 3$

This means that T_B can be 0 2 4 or 6ms based on the outcome of the random variable R.

Q) Pure ALOHA network transmit 200 bit shared channel of 200kbps. What is the requirement to make this frame collision-free?

Sol) Avg frame transmission time T_f is

$$\frac{200 \text{ bits}}{200 \text{ kbps}} = 1 \text{ ms}$$

The vulnerable time is $2 \times 1\text{ms} = 2\text{ms}$

This means no station should send later than 1ms before this station starts transmission, and no station should start during the period(1ms) that this station is sending.

Q) Suppose sender transmit 10000-bit of data & reciever recived them in 2seconds.

Sol) Throughput $= \frac{10000}{2\text{ms}} = \frac{5000}{2\text{ms}} = 5 \times 10^6 = 5000 \text{ bits/m}$

$$= \frac{\text{Amount of data}}{\text{Time taken to transmit}}$$

Q) A pure ALOHA transmit 200 bits on a shared channel of 200kbps. What is throughput if the system produces

a). 1000 frames per second
b). 500 fps
c). 250 fps

Sol) frame transmission time 200/200kbps or 1ms

(a) 1000 frames per second, then 1 frame per milisecond

$G = 1 \quad S = G \cdot e^{-2G} = (1) \times e^{-2(1)}$

$e = 2.718$

$S = 0.135 \ (13.5\%)$

(b) $1000 \times 0.155 = 135$ frames out of 1000 reaches the destination

(b) $500 \longrightarrow 1/2$ per millisecond

$G = 1/2 \quad S = G \times e^{-2G} = 0.184$

$500 \times 0.184 = 92$ of 500

(c) $250 \longrightarrow 1/4$ per millisecond

$G = 1/4 \quad S = G \times e^{-2G} = 0.152$

$200 \times 0.152 = 38$ of 250 reach the destination

Q. A network using CSMA/CD has a bandwidth of 10mbps. If the maximum propagation time (including the delays in the devices & ignoring the time needed to send a jamming signal, as we see later) is 25.6 μs, what is the minimum size of the frame.

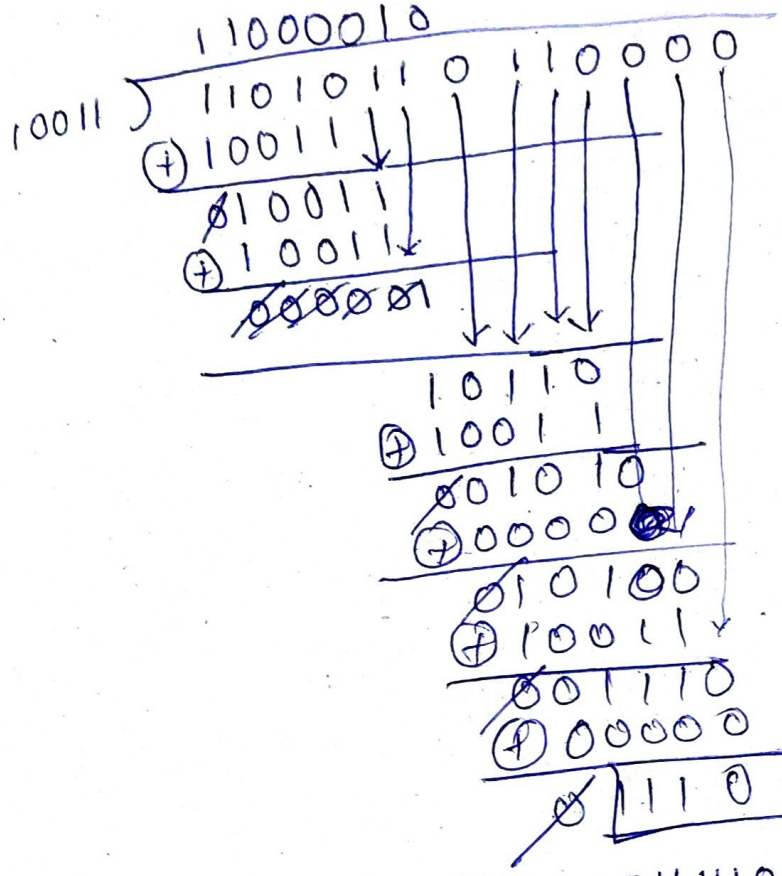
Any Minimum frame transmission time is

$T_{fr} = 2 \times T_p = 51.2\ \mu s \qquad T_p = 25.6$

This means in the worst case a station needs to transmit for a period of 51.2 μs to detect the collision.

* The Minimum size of the frame is 10mbps × 51.2 μs = 512 bits or 64 bytes. which is actually the minimum size of the frame for standard ethernet

dataword = 11010 11011 ← 4 Zeros
generator = 10011 → 4 Zeros



Codeword = 11010110111110

Codataword 1001 divisor x^3+x+1
3 2 1 0
$(x^3+1)\,x^3$ 1011

$$x^3+x$$
$$x^3+x+1 \;)\; x^6 + x^3$$
$$+\;\; x^6+x^4+x^3$$
$$+\;\; x^4$$
$$+\;\; x^4+x^2+x$$
$$x^2+x \leftarrow 110 \text{ crc}$$

Codeword → 1001110

